

Lecture 10: Relevance feedback, implicit feedback and click model

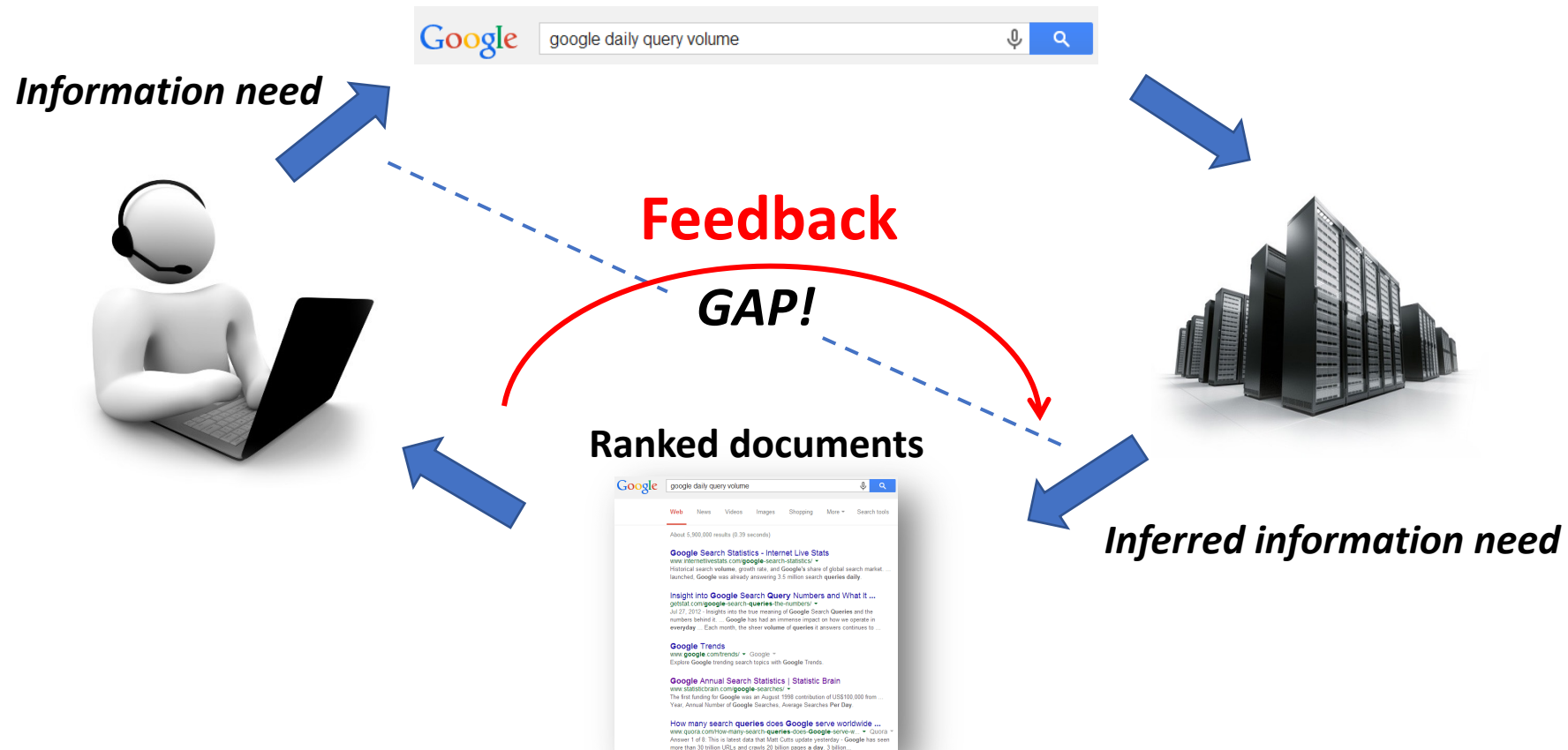
Huazheng Wang
EECS@OSU

User feedback

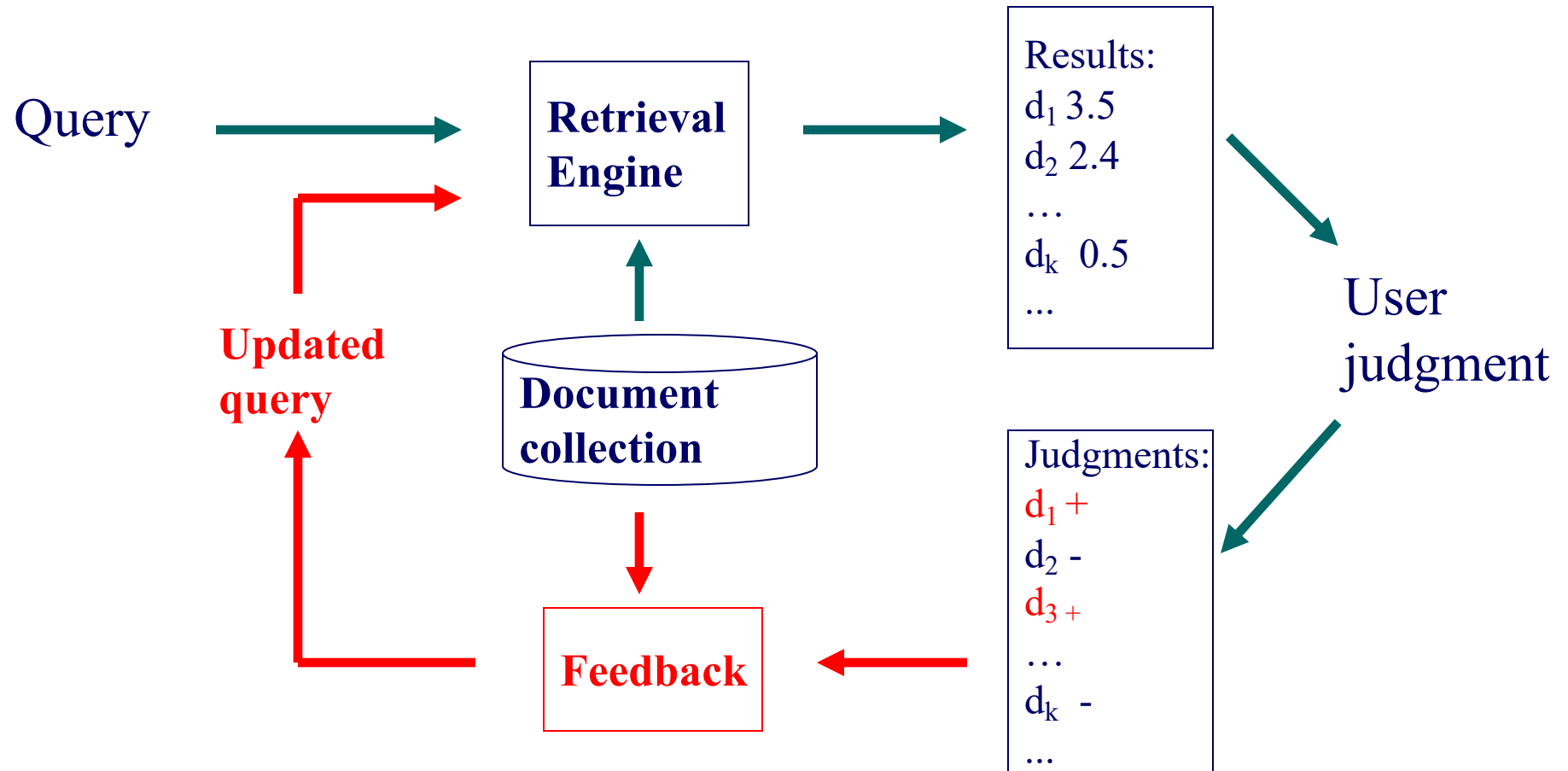
should be

- An IR system ~~is~~ an interactive system

Query



Relevance feedback



Basic idea in feedback

- Query expansion
 - Feedback documents can help discover related query terms
 - E.g., query=“information retrieval”
 - Relevant docs may likely share very related words, such as “search”, “search engine”, “ranking”, “query”
 - Expand the original query with such words will increase recall and sometimes also precision

Basic idea in feedback

- Learning-based retrieval
 - Feedback documents can be treated as supervision for ranking model update
 - Covered in the lecture of “learning-to-rank”

Relevance feedback in real systems

- Google used to provide such functions

[Personalization](#) - Wikipedia, the free encyclopedia  

Personalization involves using technology to accommodate the differences between individuals. Once confined mainly to the Web, it is increasingly becoming a ...

[en.wikipedia.org/wiki/Personalized](#) - 42k - [Cached](#) - [Similar pages](#) - 

Relevant

[Personalized Gifts from Personalization Mall](#)  

It shows you went out of your way to find the perfect gift and to **personalize** it to make it theirs alone! At PersonalizationMall.com, we design most of our ...

[www.personalizationmall.com/Default.aspx?&did=111028](#) - 47k -

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Nonrelevant

[What is personalization?](#) - a definition from Whatis.com  

Mar 6, 2007 ... On a Web site, **personalization** is the process of tailoring pages to individual users' characteristics or preferences.

[searchcrm.techtarget.com/sDefinition/0,,sid11_gci532341,00.html](#) - 72k -

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- Guess why?



what is information retrieval



Information retrieval (IR) is the process of searching for and retrieving information from a large collection of unstructured or semi-structured data, such as text documents, images, audio files, and video files. The goal of information retrieval is to provide relevant and useful information to users who are searching for it.



Relevance feedback in real systems

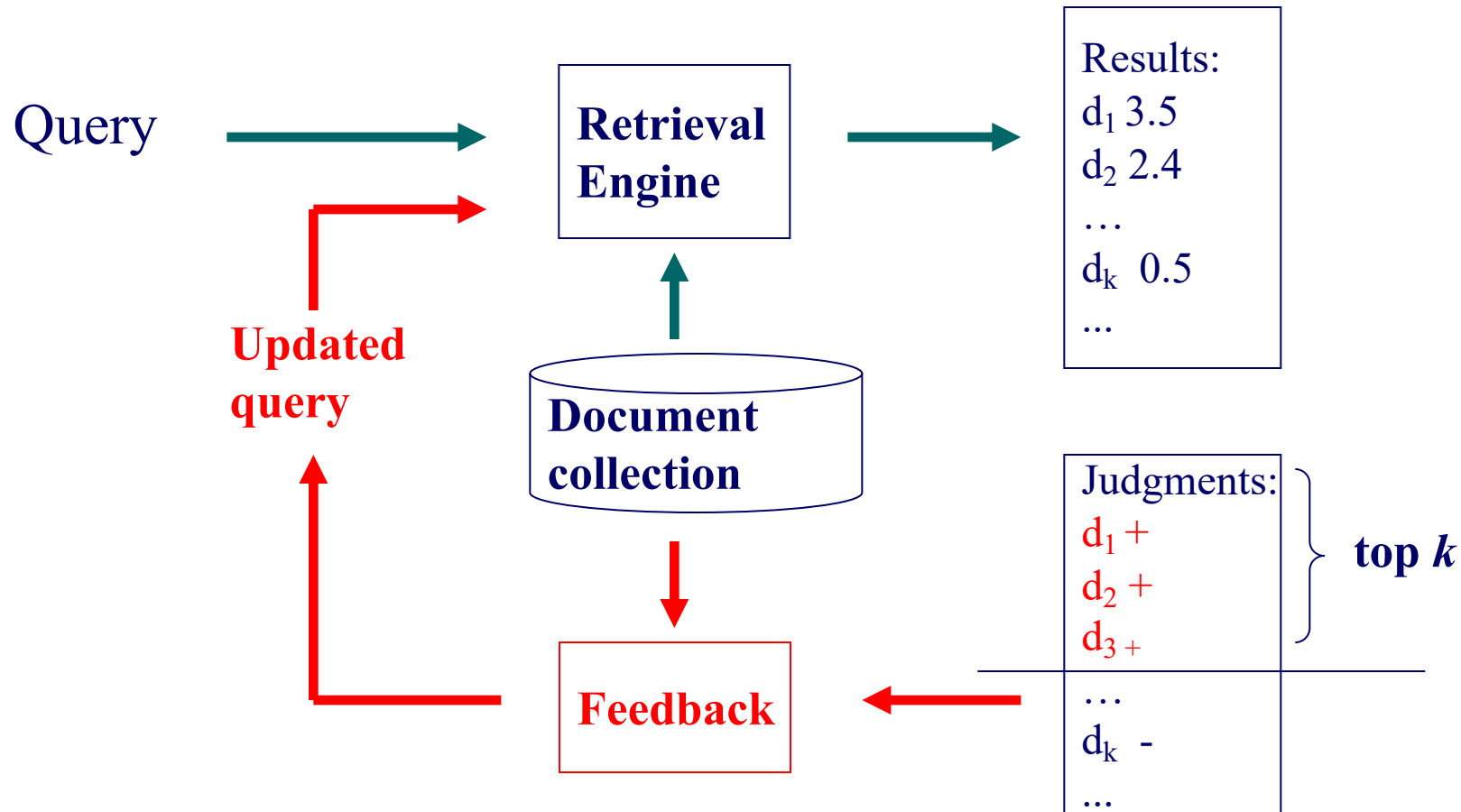
- Popularly used in image search systems

Result:

 Similarity: 1.387633 Query Image top	 Similarity: 0.440483 neutral ▾ top	 Similarity: 0.352732 neutral ▾ top	 Similarity: 0.346222 neutral ▾ top	 Similarity: 0.345664 neutral ▾ top
 Similarity: 0.340732 neutral ▾ top	 Similarity: 0.332161 neutral ▾ top	 Similarity: 0.329942 neutral ▾ top	 Similarity: 0.325042 neutral ▾ top	 Similarity: 0.323497 neutral ▾ top

Pseudo relevance feedback

- What if the users are reluctant to provide any feedback



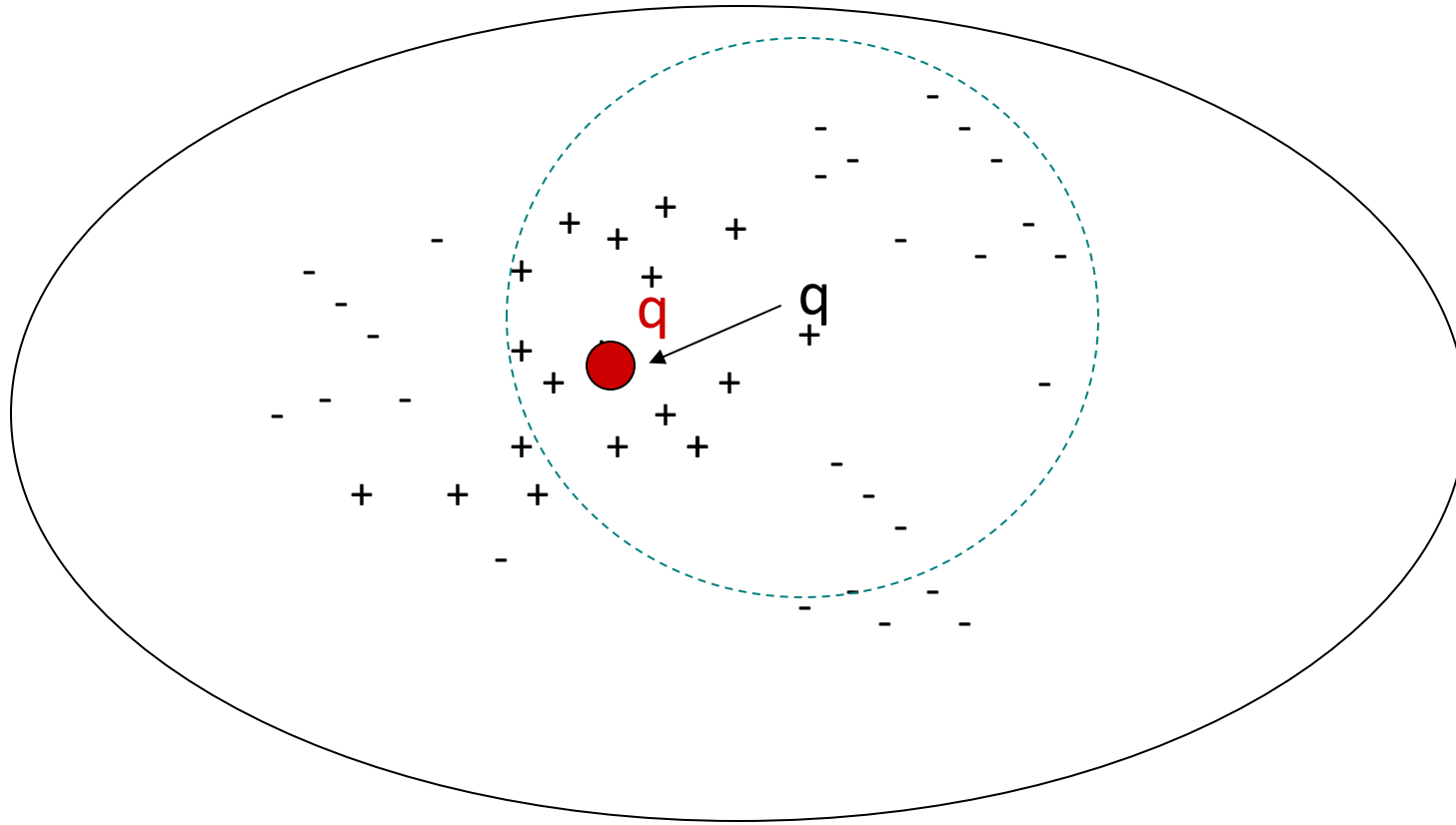
Feedback techniques

- Feedback as query expansion
 - Step 1: Term selection
 - Step 2: Query expansion
 - Step 3: Query term re-weighting
- Feedback as training signal
 - Covered in learning to rank

Relevance feedback in vector space models

- General idea: query modification
 - Adding new (weighted) terms
 - Adjusting weights of old terms
- The most well-known and effective approach is Rocchio [Rocchio 1971]
 - Find a new query that maximizes similarity with relevant docs while minimizing similarity with non relevant docs

Illustration of Rocchio feedback



Formula for Rocchio feedback

- Standard operation in vector space

The diagram illustrates the Rocchio feedback formula with the following components and annotations:

- Modified query**: Points to the resulting vector $\vec{q}_m$.
- Original query**: Points to the vector $\vec{q}$.
- Parameters**: Points to the coefficients α , β , and γ .
- Rel docs**: Points to the set of relevant documents $\forall \vec{d}_i \in D_r$.
- Non-rel docs**: Points to the set of non-relevant documents $\forall \vec{d}_j \in D_n$.

$$\vec{q}_m = \alpha \vec{q} + \frac{\beta}{|D_r|} \sum_{\forall \vec{d}_i \in D_r} \vec{d}_i - \frac{\gamma}{|D_n|} \sum_{\forall \vec{d}_j \in D_n} \vec{d}_j$$

Rocchio in practice

- Negative (non-relevant) examples are not very important (why?)
- Efficiency concern
 - Restrict the vector onto a lower dimension (i.e., only consider highly weighted words in the centroid vector)
- Avoid “training bias”
 - Keep relatively high weight on the original query
- Can be used for relevance feedback and pseudo feedback
- Usually robust and effective

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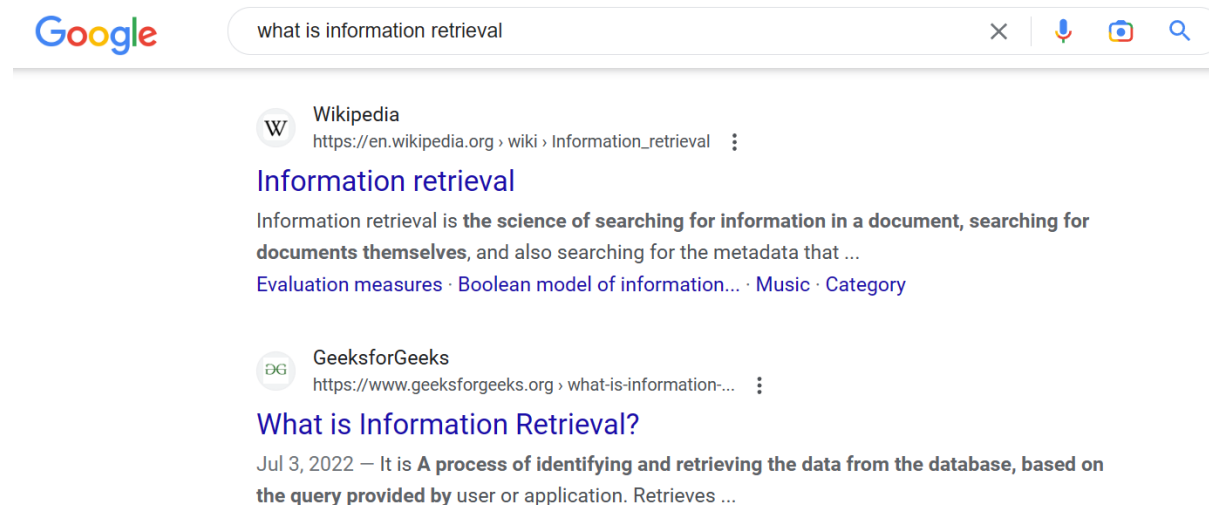
[searchcrm.techtarget.com/sDefinition/0,,sid11_gci532341,00.html](#) - 72k -

[Cached](#) - [Similar pages](#) - 

- **Vulnerable** to spammers

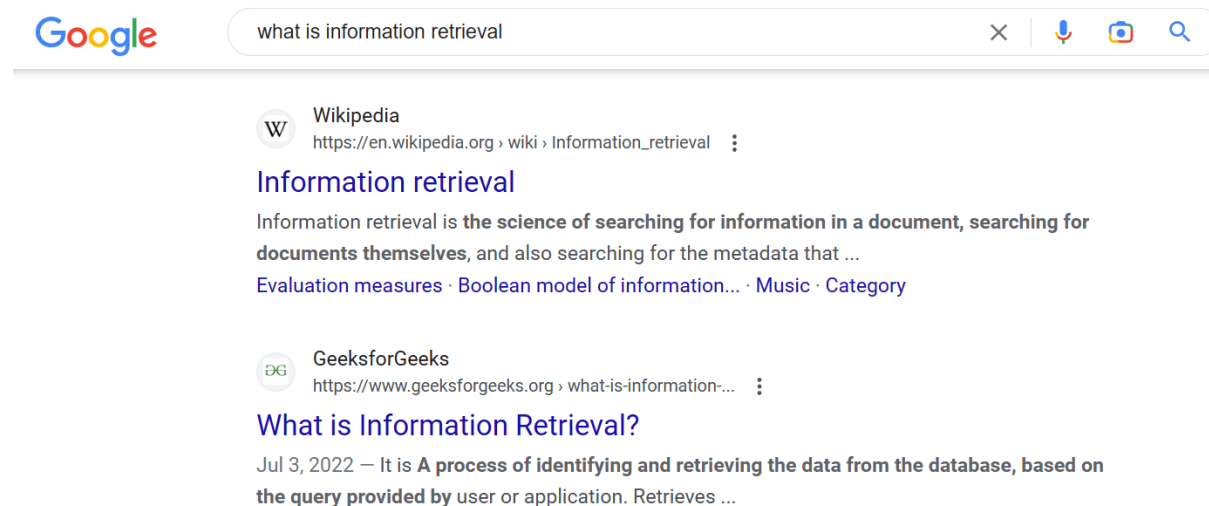
How about using clicks

- Clicked document as relevant, non-clicked as non-relevant
 - Cheap, largely available



Is click reliable?

- Why do we click on the returned document?
 - Title/snippet looks attractive
 - We haven't read the full text content of the document
 - It was ranked higher
 - Belief bias towards ranking
 - We know it is the answer!



Is click reliable?

- Why do not we click on the returned document?
 - Title/snippet has already provided the answer
 - Instant answers, knowledge graph
 - Extra effort of scrolling down the result page
 - The expected loss is larger than skipping the document
 - We did not see it....

Can we trust click as relevance feedback?



Accurately Interpreting Clickthrough Data as Implicit Feedback [Joachims SIGIR'05]

- Eye tracking, click and manual relevance judgment to answer
 - Do users scan the results from top to bottom?
 - How many abstracts do they read before clicking?
 - How does their behavior change, if search results are artificially manipulated?

Which links do users view and click?

- Positional bias

Fixations: a spatially stable gaze lasting for approximately 200-300ms, indicating visual attention

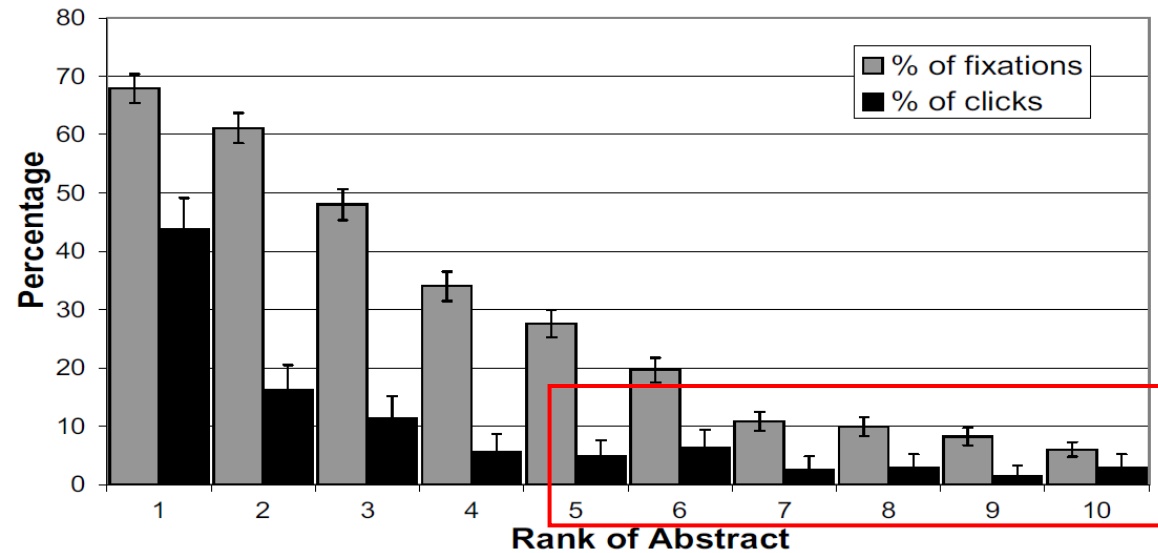


Figure 1: Percentage of times an abstract was viewed/clicked depending on the rank of the result.

Do users scan links from top to bottom?

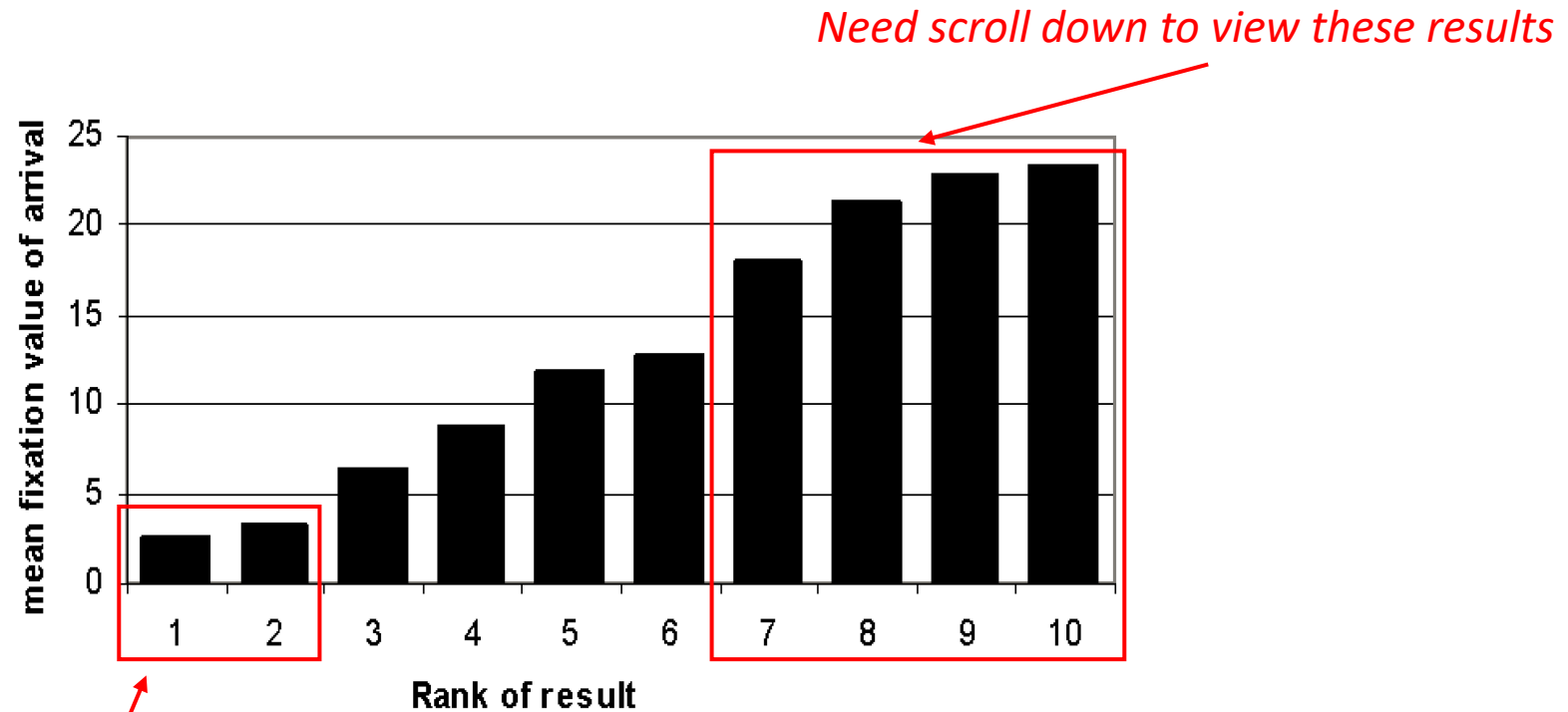


Figure 2: Mean time of arrival (in number of previous fixations) depending on the rank of the result.

View the top two results within the second or third fixation

Which links do users evaluate before clicking?

- The lower the click in the ranking, the more abstracts are viewed before the click

Table 2: Percentage of times the user viewed an abstract at a particular rank before he clicked on a link at a particular rank.

Viewed Rank	Clicked Rank					
	1	2	3	4	5	6
1	90.6%	76.2%	73.9%	60.0%	54.5%	45.5%
2	56.8%	90.5%	82.6%	53.3%	63.6%	54.5%
3	30.2%	47.6%	95.7%	80.0%	81.8%	45.5%
4	17.3%	19.0%	47.8%	93.3%	63.6%	45.5%
5	8.6%	14.3%	21.7%	53.3%	100.0%	72.7%
6	4.3%	4.8%	8.7%	33.3%	18.2%	81.8%

Does relevance influence user decisions?

- Controlled relevance quality
 - Reverse the ranking from search engine
- Users' reactions
 - Scan significantly more abstracts than before
 - Less likely to click on the first result
 - Average clicked rank position drops from 2.66 to 4.03
 - Average clicks per query drops from 0.8 to 0.64

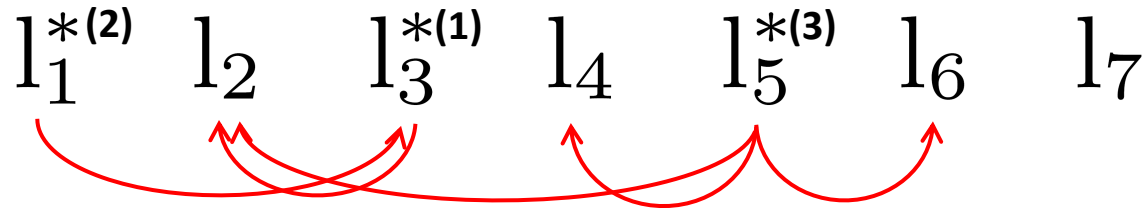
Are clicks absolute relevance judgments?

- Position bias
 - Focus on position one and two, equally likely to be viewed

“normal”	l_1^-, l_2^-	l_1^+, l_2^-	l_1^-, l_2^+	l_1^+, l_2^+	total
$\text{rel}(l_1) > \text{rel}(l_2)$	15	19	1	1	36
$\text{rel}(l_1) < \text{rel}(l_2)$	11	5	2	2	20
$\text{rel}(l_1) = \text{rel}(l_2)$	19	9	1	0	29
total	45	33	4	3	85

Are clicks relative relevance judgments?

- Clicks as pairwise preference statements
 - Given a ranked list and user clicks



- Click > Skip Above
- Last Click > Skip Above
- Click > Earlier Click
- Last Click > Skip Previous
- Click > Skip Next

Clicks as pairwise preference statements

- Accuracy against manual relevance judgment

Explicit Feedback Data Strategy	Abstracts				
	Phase I “normal”	“normal”	Phase II “swapped”	“reversed”	all
Inter-Judge Agreement	89.5	N/A	N/A	N/A	82.5
Click > Skip Above	80.8 ± 3.6	88.0 ± 9.5	79.6 ± 8.9	83.0 ± 6.7	83.1 ± 4.4
Last Click > Skip Above	83.1 ± 3.8	89.7 ± 9.8	77.9 ± 9.9	84.6 ± 6.9	83.8 ± 4.6
Click > Earlier Click	67.2 ± 12.3	75.0 ± 25.8	36.8 ± 22.9	28.6 ± 27.5	46.9 ± 13.9
Click > Skip Previous	82.3 ± 7.3	88.9 ± 24.1	80.0 ± 18.0	79.5 ± 15.4	81.6 ± 9.5
Click > No Click Next	84.1 ± 4.9	75.6 ± 14.5	66.7 ± 13.1	70.9 ± 15.7	70.4 ± 8.0

How accurately do clicks correspond to explicit judgment of a document?

- Accuracy against manual relevance judgment

Explicit Feedback Data Strategy	Pages Phase II all
Inter-Judge Agreement	86.4
Click > Skip Above	78.2 $\pm$ 5.6
Last Click > Skip Above	80.9 $\pm$ 5.1
Click > Earlier Click	64.3 $\pm$ 15.4
Click > Skip Previous	80.7 $\pm$ 9.6
Click > No Click Next	67.4 $\pm$ 8.2

What do we get from this user study?

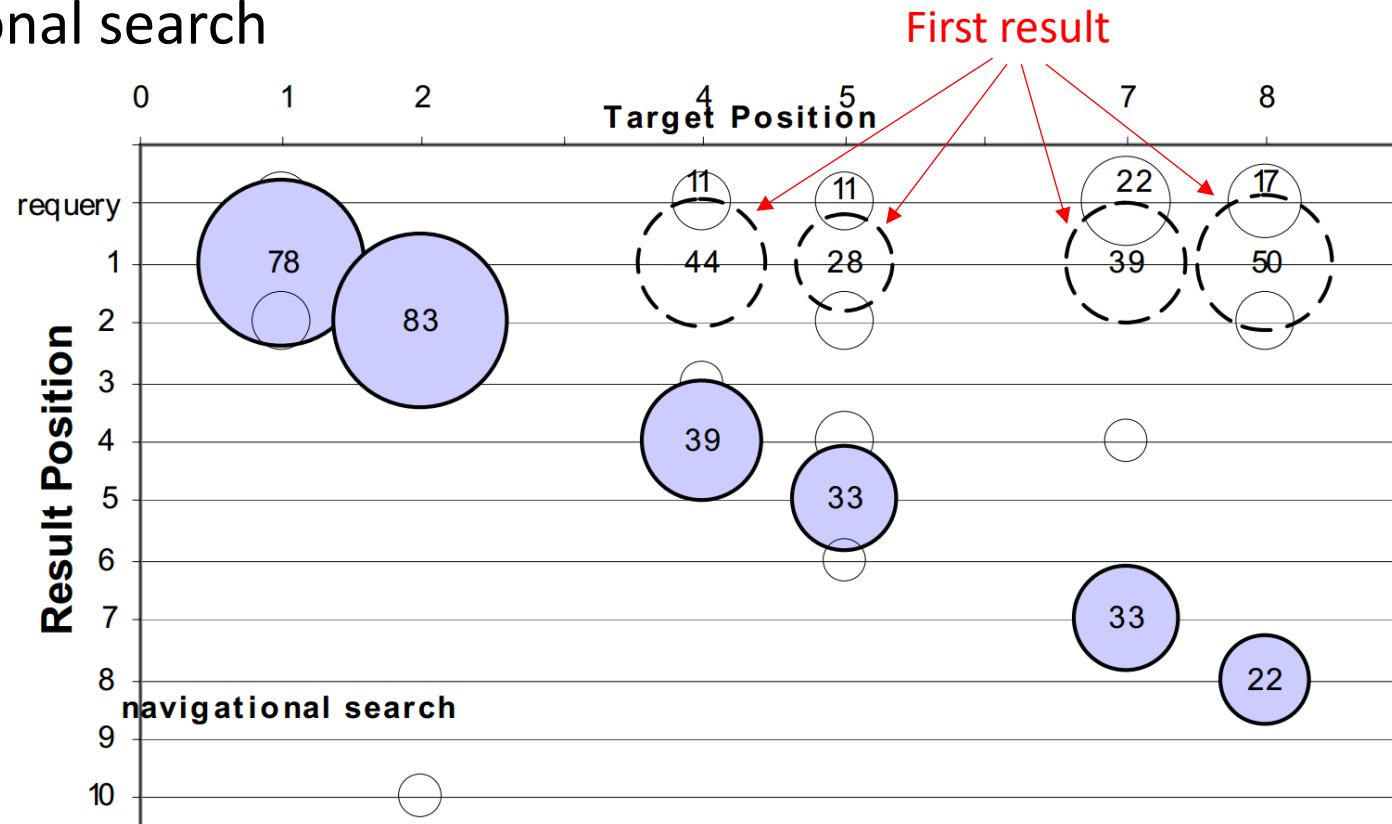
- Clicks are influenced by the relevance of results
 - Biased by the trust over rank positions
- Clicks as relative preference statement is more accurate
 - Several heuristics to generate the preference pairs

How to utilize such preference pairs?

- Pairwise learning to rank algorithms
 - We have covered it in learning to rank discussions
 - Recall: LambdaRank / LambdaMART

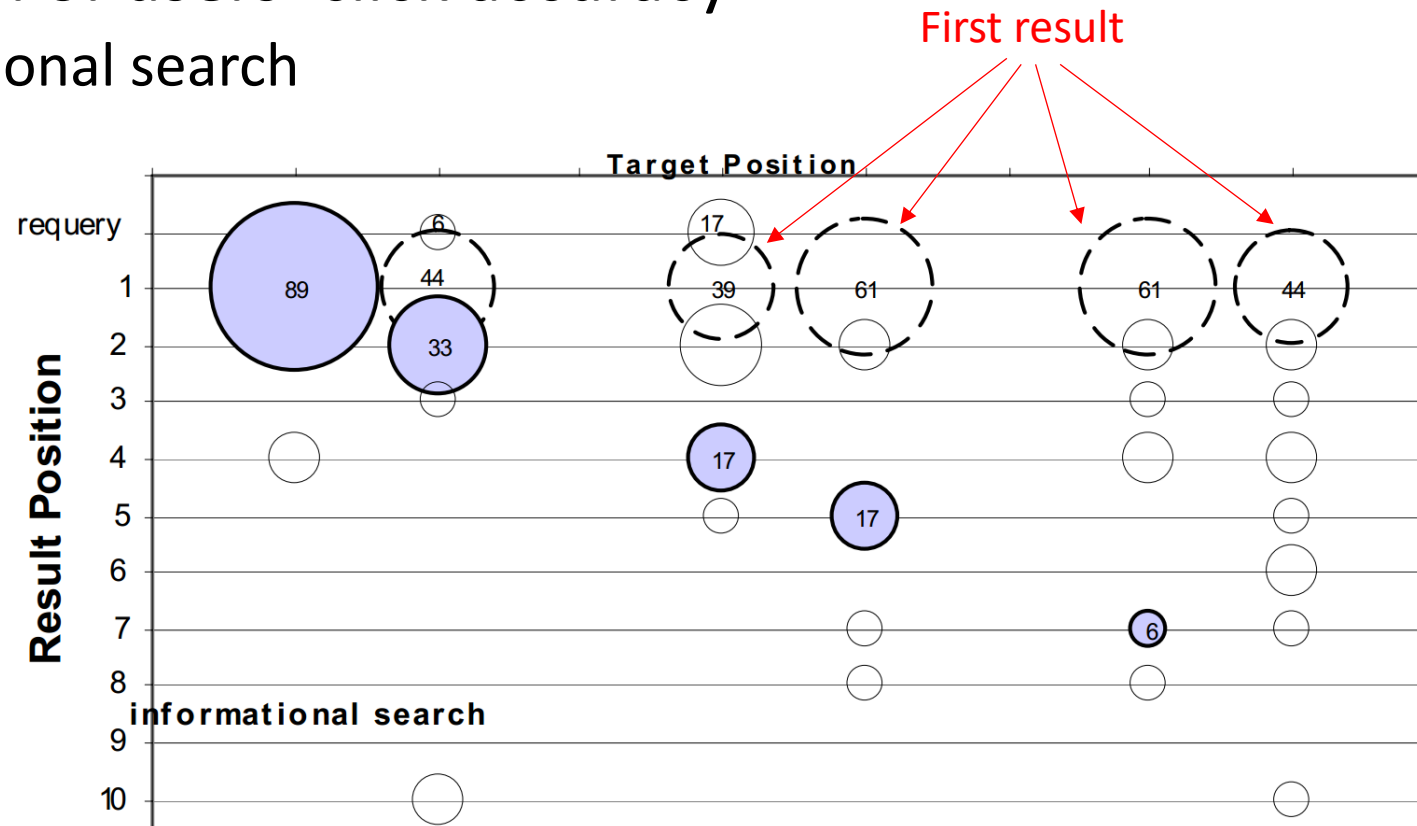
An eye tracking study of the effect of target rank on web search [Guan CHI'07]

- Break down of users' click accuracy
 - Navigational search



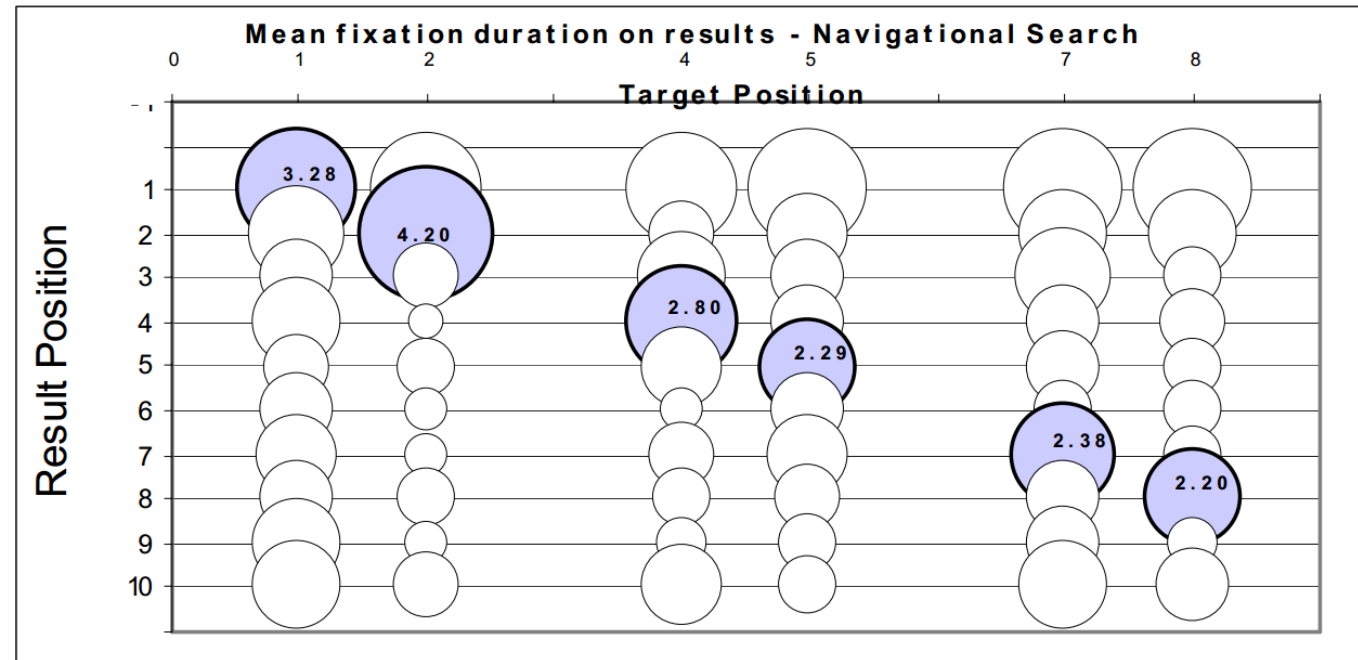
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- Break down of users' click accuracy
 - Informational search



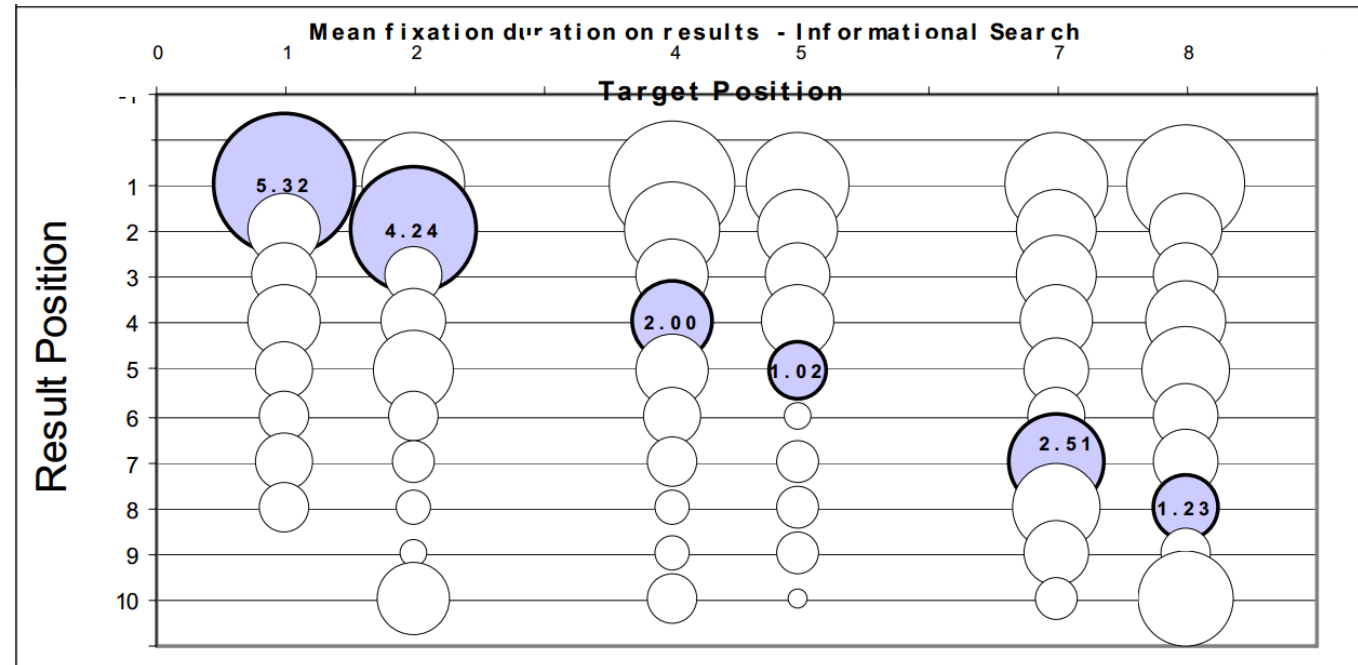
Users failed to recognize the target because they did not read it!

- Navigational search



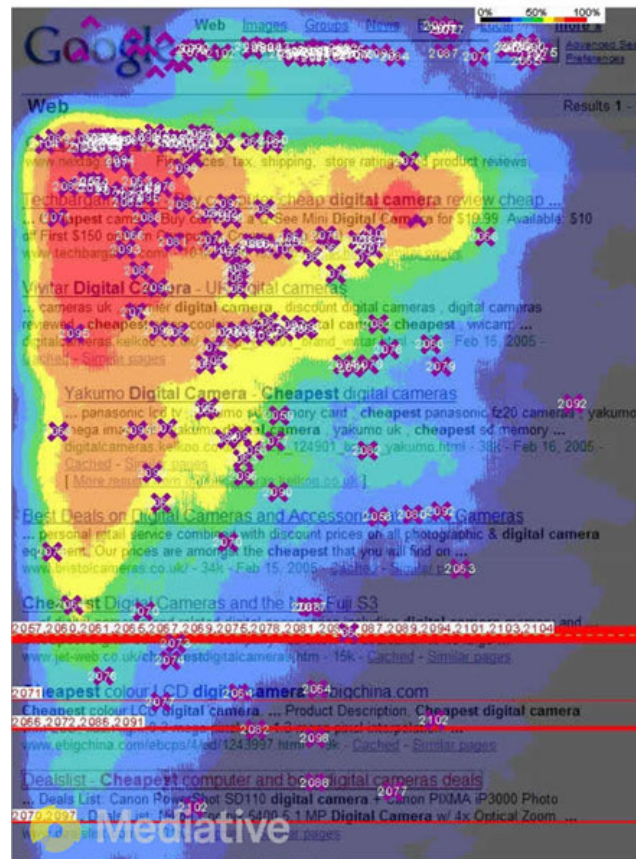
Users did not click because they did not read the results!

- Informational search

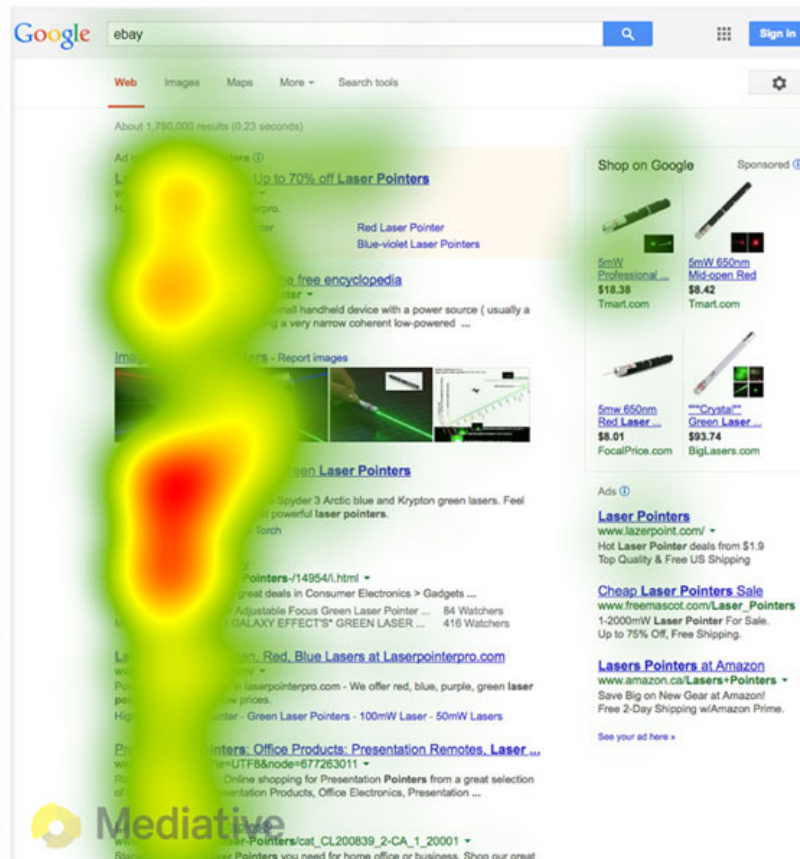


Where do we pay attention to?

2005



2014



Predicting clicks: estimating the click-through rate for new ads [Richardson WWW'07]

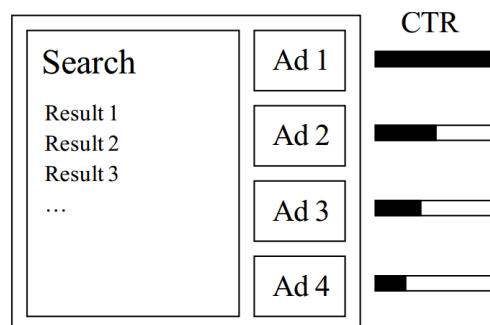
- To maximize ad revenue

- $E_{ad}[Revenue] = \sum_{ad} p(click|ad)CPC(ad)$

Estimated click-through rate

Cost per click: basic business model in search engines

- Position-bias is also true in online ads
 - Observed low CTR is not just because of ads' quality, but also their displayed positions!



Combat position-bias by explicitly modeling it

- Being clicked is related to its quality and position
 - $p(click|ad, pos) = p(click|ad, pos, seen)p(seen|pos)$

$$= \underline{p(click|ad, seen)} \underline{p(seen|pos)}$$

Calibrated CTR for ads ranking

Discounting factor

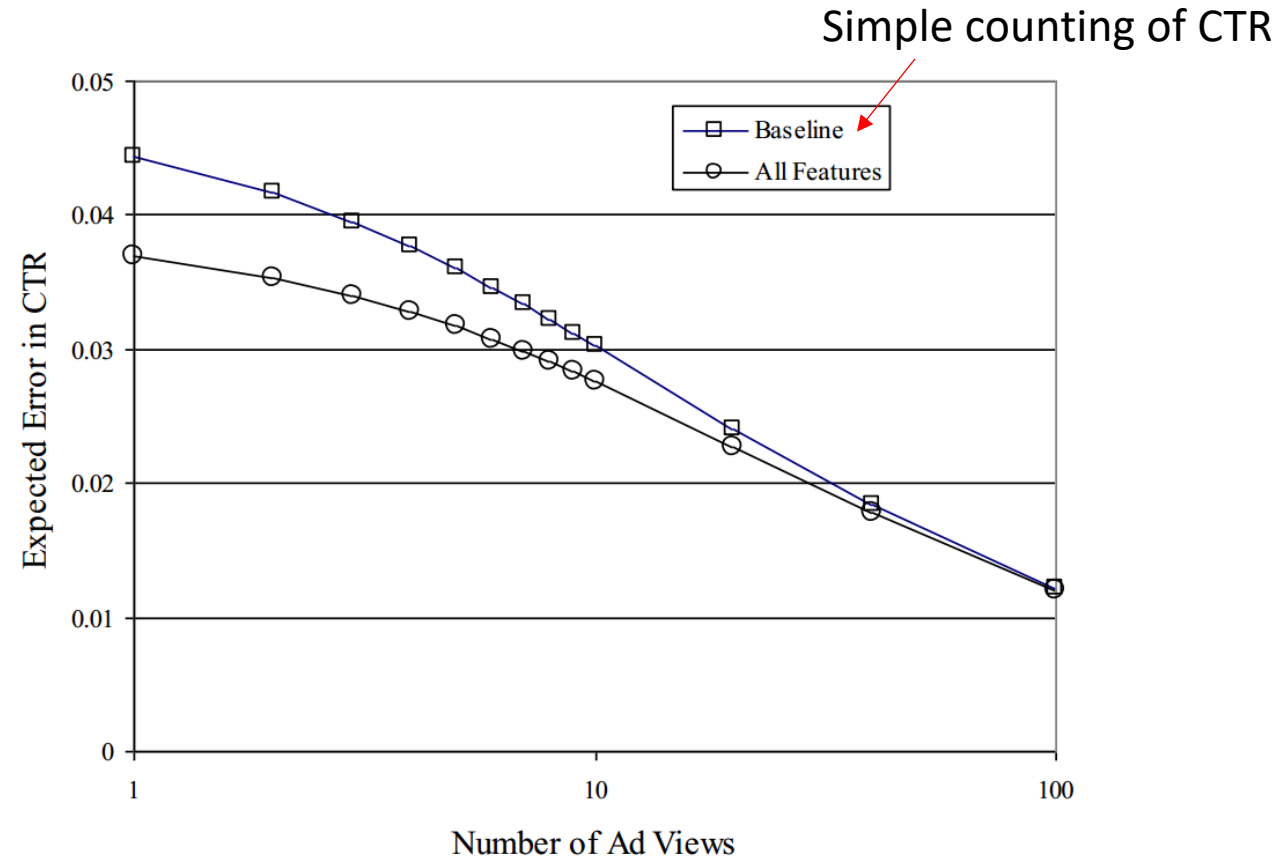
- $p(click = 1|ad, seen = 0) = 0$
- $p(click = 1|ad, seen = 1) = \frac{1}{1 + \exp(-w^T f_{ad})}$

Logistic regression by features of the ad

Parameter estimation

- Discounting factor
 - Approximation: positions being clicked must be seen already
 - $p(\textit{seen}|\textit{pos}) \propto \#\textit{clicks_at_pos}$
- Calibrated CTR
 - Maximum likelihood for w with historic clicks
 - $\hat{w} = \operatorname{argmax}_w \sum_{ad} \log p(\textit{click}|ad, pos)$

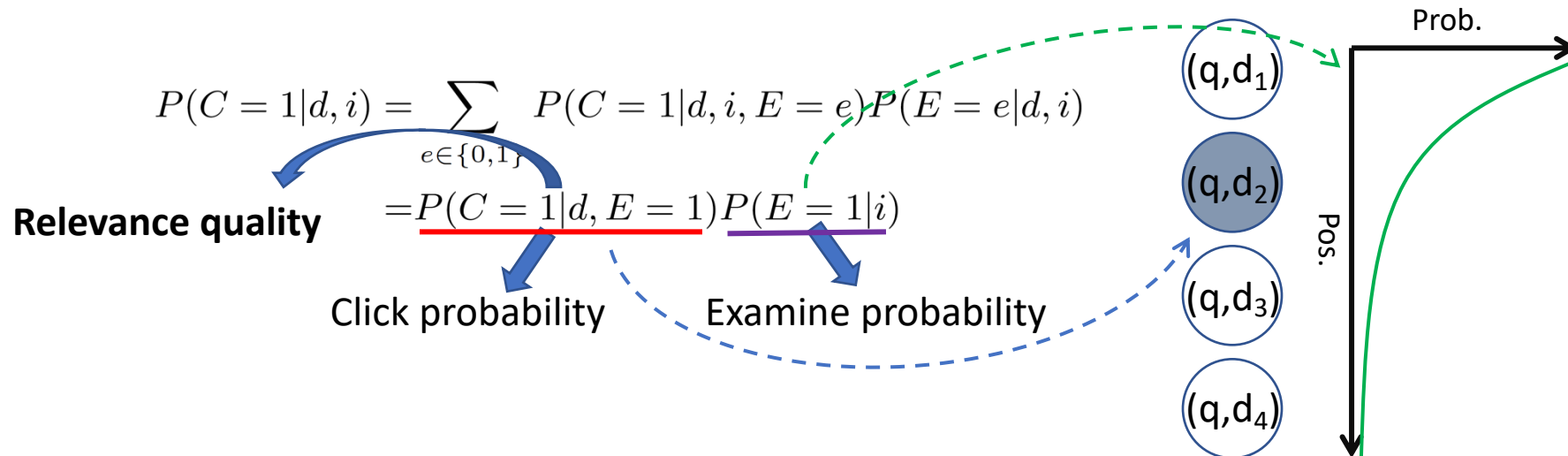
Calibrated CTR is more accurate for new ads



- Unfortunately, their evaluation criterion is still based on biased clicks in testing set

Click models

- Decompose relevance-driven skips from position-driven skips
 - Examine: user reads the displayed result
 - Click: user clicks on the displayed result
 - Atomic unit: (query, doc)
- Position-based model (PBM)



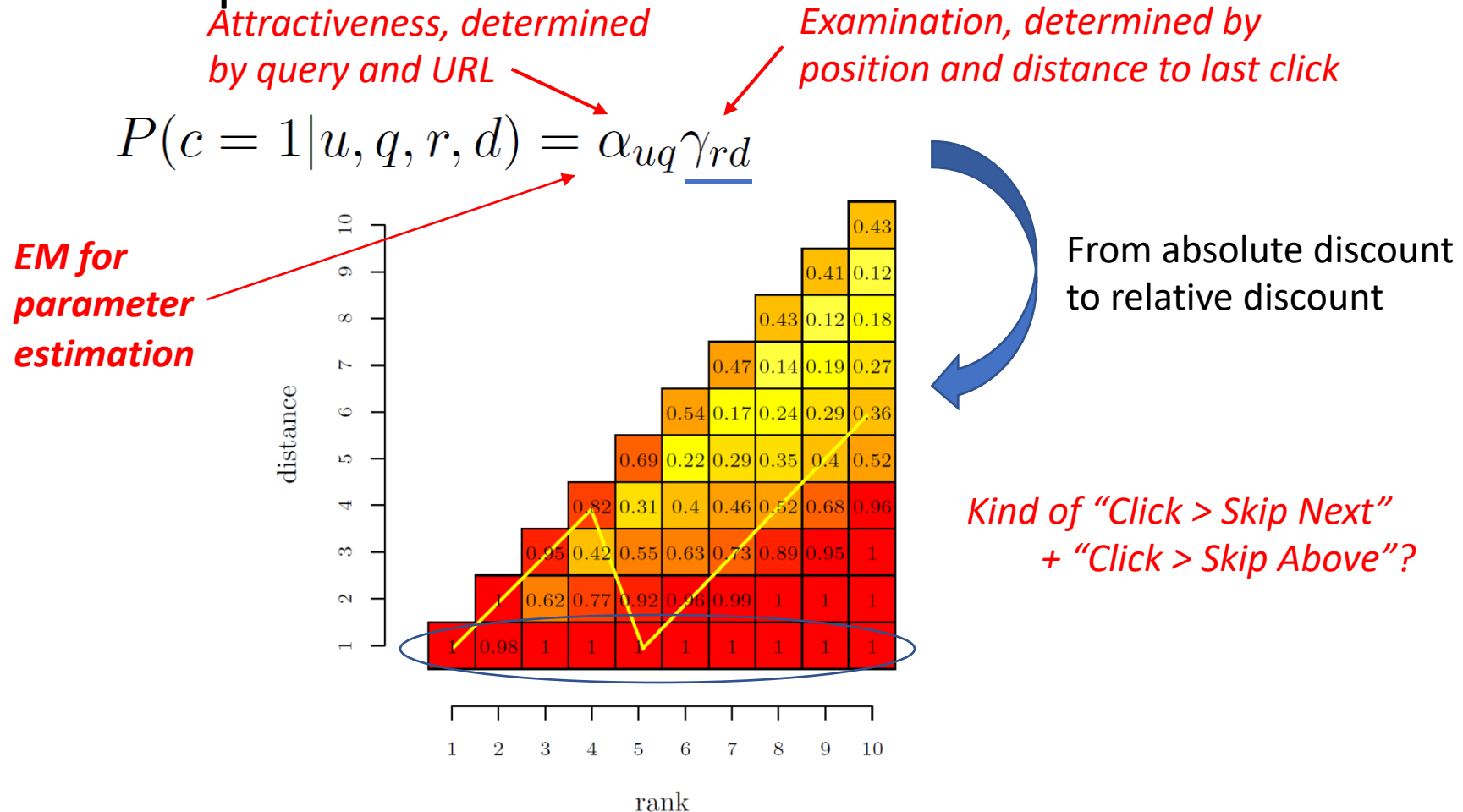
Cascade Model [Craswell et al. WSDM'08]

- Sequential browsing assumption
 - At each position decides whether to move on
 - $p(C_i = 1) = p(R_i = 1) \prod_{j=1}^{i-1} (1 - p(R_j = 1))$
 - Assuming relevance $R_i = 1 \rightarrow C_i = 1$
 - Only one click is allowed on each search result page

Kind of "Click > Skip Above"?

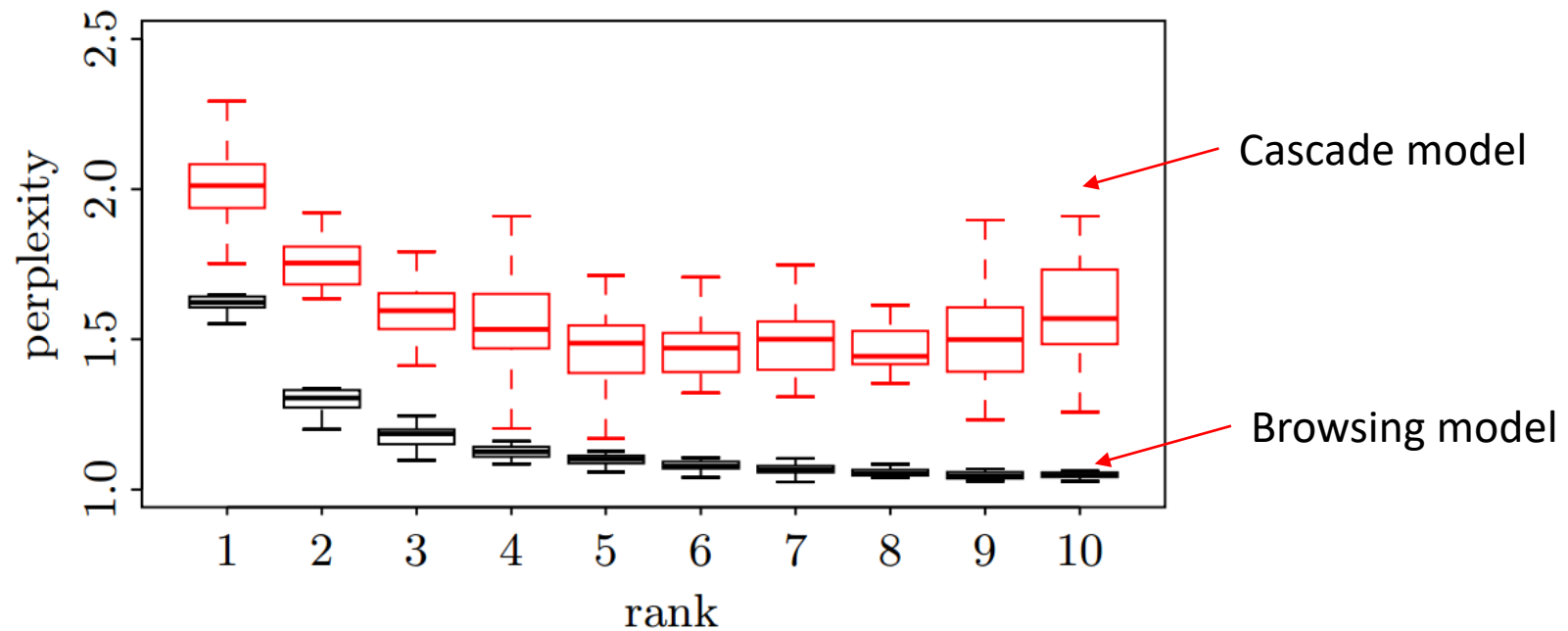
User Browsing Model [Dupret et al. SIGIR'08]

- Examination depends on distance to the last click



More accurate prediction of clicks

- Perplexity – randomness of prediction



Revisit User Click Behaviors

The screenshot shows a Yahoo! News search results page for the query "jeremy lin". The page includes a search bar, navigation tabs (WEB, IMAGES, VIDEO, NEWS, MORE), and filter options (FILTER BY SOURCE, FILTER BY TIME, RELATED CONCEPTS). The search results are sorted by Relevance. Several articles are listed, with red dotted circles highlighting specific titles and snippets. A yellow smiley face icon is positioned next to the first article, with three thought bubbles containing red text. A vertical dashed red line with arrows points from the first article to the second, and another points from the second to the third. A checkbox is next to the first article, and a box with an 'X' is next to the second article.

YAHOO! NEWS jeremy lin Search 10,181 results

WEB IMAGES VIDEO **NEWS** MORE

Also try: [jeremy lin knicks](#), [jeremy lin shoe](#), [More...](#)

Sort Results by: Relevance | Time

Knicks-Pistons Linsanity Live Stream: Where To Watch Jeremy Lin For Free Online
Can Jeremy Lin and the New York Knicks get back to their winning ways under new coach Mike Woodson? We'll find out Saturday night as they host the lowly Detroit Pistons at Madison Square Garden.
International Business Times - Mar 24 02:08pm

Bats: Less of a Frenzy for Lin in Toronto This Time
Jeremy Lin's second trip to Toronto with the Knicks has been a less intense than his first.
New York Times - Mar 23 10:58am

Off the Dribble: Less of a Frenzy for Lin in Toronto This Tim... - New York Times

Less of a Frenzy for Lin in Toronto This Time - New York Times

Grange on NBA: The life of Jeremy Lin - Sportsnet.ca

[all 142 news articles...](#)

Jeremy Lin Update: Knicks Win. Lin Still Plagued by Turnovers | The Harvard Crimson
Jeremy Lin and the Knicks got back on track at Madison Square Garden on Saturday against the Detroit Pistons. The Pistons stayed within close range for most of the first three quarters, but New York pulled away in the fourth to cruise to a 101-79 victory.
The Harvard Crimson - Mar 24 11:35pm

New Coach Runs Similar Show, and Lin Stays in Picture - New York Times

Jeremy Lin's Brand Thriving With Knicks New Coach - Forbes

NBA: Rajon Rondo prepared for showdown with Jer... - Worcester Telegram & Gazette

[all 14 news articles...](#)

Match my query?

Redundant doc?

Shall I move on?

Summary

- Purpose of relevance feedback
- Rocchio relevance feedback for vector space models
- Implicit feedback
- Click models